CRYPTOGRAPHY - 6

1. E = 4

   V = 21

   I = 8

   R = 17

   Decryption : P = (C-K) mod 26

K = 0	Plaintext : EVIRE	K = 16	P: OFSBO
K = 1	Plaintext : DUHQD	K = 17	P: NERAN
K = 2	P: CTGPC	K = 18	P: MDQ2M
K = 3	P: BSFOB	K = 19	P: LCPYL
K = 4	P: ARENA	K = 20	P: KBOXK
K = 5	P: 2QDM2	K = 21	P: JANWJ
K = 6	P: YPCLY	K = 22	P: I2MVI
K = 7	P: XOBKX	K = 23	P: HYLUH
K = 8	P: WNAJW	K = 24	P: GXKTG
K = 9	P: VM2IV	K = 25	P: FWJSF
K = 10	P: ULYHU		
K = 11	P: TKXGT		
K = 12	P: SIWFS		
K = 13	P: RIVER		
K = 14	P: QHUDQ		
K = 15	P: PGTCP		

a) Meaningful plaintexts

K = 13

. E (4 - 13) mod 26 : -9 mod 26

= 17 → R

V (21 - 13) mod 26

= 8 → I

I (8 - 13) mod 26 : -5 mod 26

= 21 → V

R (17 - 13) mod 26

= 4 → E

E (4 - 13) mod 26 = -9 mod 26

= 17 → R

RIVER

K = 4

E = (4 - 4) mod 26

= 0 → A

V = (21 - 4) mod 26

= 17 → R

I = (8 - 4) mod 26

= 4 → E

R = (17 - 4) mod 26

= 13 → N

E = (34 - 4) mod 26

= 0 → A

ARENA

b).

Keys give 2 meaningful texts

ie RIVER and ARENA. since it is mentioned either at ticket level of colosseum so it could mean RIVER

c) A ciphertext only attack can produce more than one meaningful plaintext. When ciphertext is short & Key space is small, the correct key is not uniquely determined. The attacker needs more ciphertext or extra context to decide

2. Plaintext → Ciphertext

HAHAHA → NONONO

a.

H → 7 N → 13

A → 0 O → 14.

b. $E(X) = ax + b \pmod{26}$

$7a + b \equiv 13 \pmod{26}$

$0 \cdot a + b \equiv 14 \pmod{26}$

c. $7a + b = 13 \bmod 26$

$0 + b \equiv 14 \bmod 26$

$\Rightarrow b = 14 \bmod 26$

$b = 14.$

Substitute value of b

$7a + 14 \equiv 13 \bmod 26$

$7a = 13 - 14 \pmod{26}$

$7a = -1 \bmod 26$

$7a = 25 \bmod 26$

Inverse of $7^{-1} = 15$

$a = 25 \times 15 = 375$

$= 14 \times 26 + 11$

$a = 11$

$E(x) = (11x + 14) \bmod 26.$

3. CRWW2 CR → HA

a) $H = 7 \quad A = 0$

$C = 2 \quad R = 17$

$7a + b \equiv 2 \bmod 26$

$0a + b = 17 \bmod 26$

$b = 17 \bmod 26 = 17$

substitute $b = 17$

$7a + 17 \equiv 2 \bmod 26$

$7a \equiv -15 \equiv 11 \bmod 26$

$a \equiv 11 \times 15 = 165 = 6 \times 26 + 9$

$a = 9$

b) $E(x) = (9x + 17) \bmod 26$

c) $9^{-1} \bmod 26$

$9 \times 3 = 27 = 26 + 1$

$9^{-1} = 3$

$D(Y) = 3(Y - 17) \bmod 26$

$= (3Y - 51) \bmod 26$

since $-51 + 52 = 1$

$D(Y) = (3Y + 1) \bmod 26.$

d) Decrypt CRWW2.

C = 2 R = 17, W = 22, W = 22, 2 = 25

C = $3(2) + 1 = 7 \bmod 26 = 7 \rightarrow H$

R = $3(17) + 1 = 52 \bmod 26 = 0 \rightarrow A$

W = $3(22) + 1 = 67 \bmod 26 = 15 \rightarrow P$

2 = $3(25) + 1 = 76 \bmod 26 = 24 \rightarrow Y$

HAPPY

4.

a.

RIVER R = 17 I = 8 V = 21 E = 4 R = 17

NQXTP N = 13 Q = 16 X = 23 T = 19 P = 15

b. encryption

$C = (P + K) \bmod 26$

R. $17 + 13 = 30 \bmod 26 = 4 \rightarrow E$

I $= 8 + 16 = 24 \bmod 26 = 24 \rightarrow Y$

V $= 21 + 23 = 44 \bmod 26 = 18 \rightarrow S$

E $= 4 + 19 = 23 \bmod 26 = 23 \rightarrow X$

R $= 17 + 15 = 32 \bmod 26 = 6 \rightarrow G$

$\therefore$ Ciphertext = EYSXG

c. Decryption $P = (C - K) \bmod 26$

E $= 4 - 13 = -9 \bmod 26 = 17 \rightarrow R$

Y $= 24 - 16 = 8 \bmod 26 = 8 \rightarrow I$

S $= 18 - 23 = -5 \bmod 26 = 21 \rightarrow V$

X $= 23 - 19 = 4 \bmod 26 = 4 \rightarrow E$

G $= 6 - 15 = -9 \bmod 26 = 17 \rightarrow R$

Plaintext = RIVER

d. Random - If key has any pattern the attacker can guess likely key letters & test them, so some plaintext becomes more likely than others. A truly random key makes every key letter equally likely, so every plaintext letter is equally likely too.

- Equal length to the plaintext - If key is shorter, it must repeat. Repetition creates a periodic structure that can be broken with frequency analysis, while a full length key each plaintext has its own independent key, and shows no pattern.

- used only once : If the same key K encrypts 2 messages then $C_1 - C_2 \equiv P_2 \bmod 26$ the key cancels out & leaks the relationship between 2 plaintexts. If the attacker also knows one plaintext, the key is recovered directly:
$K \equiv C - P \bmod 26$.